

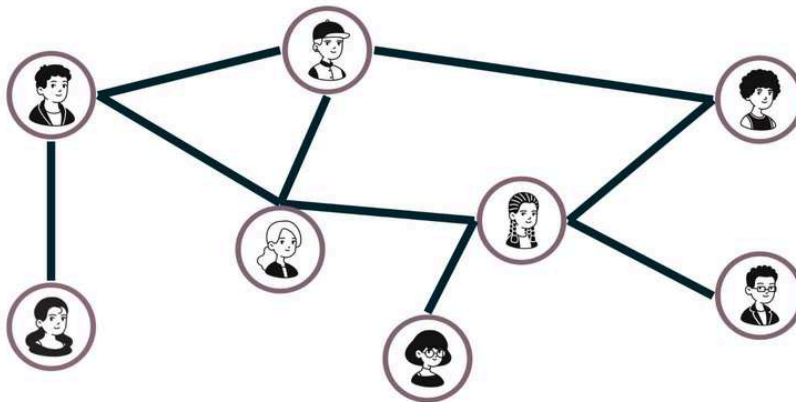
L5: What Does “Small-world” Network Mean?

A **small-world network** is a network that satisfies the small-world property. To determine if a network has small-world property, we compare two of its characteristics against an ensemble of random $G(n,p)$ networks with the same size n and density p .

For the first condition we check, *is the clustering coefficient **much larger** than the random $G(n,p)$ networks?* This condition can be examined with an appropriate hypothesis test. For instance, the one-sample one-tailed t-test would examine, in this case, whether the clustering coefficient of the given network is significantly greater than the mean of the clustering coefficient values in the $G(n,p)$ ensemble.

For the second condition, we again *check that the CPL of the given network is **not significantly greater** than the mean CPL value in the $G(n,p)$ ensemble.*

Please note that whether the previous conditions hold or not may depend on the significance level (“alpha value”) of the corresponding hypothesis tests. Additionally, it may be that even though the given network has a CPL (for instance) that is greater than the mean of the CPL values in the $G(n,p)$ ensemble, the difference may be small in absolute magnitude (e.g., 2.1 versus 2.0).



280_L5_What Does “Small-world”...
3 videos



Clustering and Efficiency in Real-world Networks 2:24



Watts Strogatz Model 2:55



Degree Distribution of ... 1:09

Clustering & Efficiency in Real-world Networks



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Food For Thought

1. Can you think of networks that have both weak clustering (similar to $G(n,p)$) and long paths (*similar to regular networks*)? Can you think of any real-world networks that may have these properties?
2. The Watts-Strogatz model described above is only one possible model to create small-world networks. Can you think of other approaches to create a network that is both highly clustered and with short paths ($CPL \sim O(\log(n))$)?
3. Look at the literature for mathematical expressions that show the transitivity or average clustering coefficient as a function of n , $\bar{k}$, and p for the Watts-Strogatz model. Similarly, look for mathematical expressions for the average path length or

diameter.

4. In Lesson-4, we briefly reviewed the preferential attachment model, which is able to produce power-law networks. How would you combine the Watts-Strogatz model with the Preferential Attachment model so that you get:
- Strong clustering
 - Short paths
 - Power-law degree distribution